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Capstone Project

Heart Disease Prediction Final Report

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This document is a comprehensive report on a Heart Disease Prediction Project using machine learning, following the CRISP-DM methodology.

1. Problem Statement

Goal: This project aims to predict the likelihood of a person having heart disease using their biometric and preliminary medical markers, such as Electrocardiogram (ECG) results, fasting blood sugar levels, and reported chest pain, thereby guiding the appropriate use of costly and invasive advanced diagnostics.

Significance: Heart disease is the principal cause of mortality globally. Early, non-invasive prediction can guide the appropriate use of costly and invasive advanced diagnostics (like angiograms) and facilitate proactive, personalized care to reduce healthcare costs.

If an individual is predicted to have underlying heart disease, advanced diagnostics, including invasive procedures and medical imaging, can be appropriately performed. This project also aims to minimize unnecessary medical interventions for those not determined to be at risk. The prediction tool facilitates early detection, enabling primary intervention through proactive, personalized care, which can reduce costs for patients and insurance providers.

Research Question: Can the likelihood of a person having heart disease be predicted based on their personal and medical parameters?

Front End User Interface: Based on the analysis, a user facing application was built using “Streamlit” that can be accessed here. The application obtains the user’s information and runs a predictive model to assess the likelihood of the person having heart disease

2. Model Outcomes or Predictions

To make the right prediction, a classification-based supervised modeling approach is used as the model tries to predict whether a person has heart disease (a binary outcome: yes or no). The model also uses a set of supervised learning algorithms including Logistic Regression, Random Forest Classifier, and Support Vector Machine. The accuracy of the model is defined by calculating the F1 score, as a high F1-score indicates that the model has good performance in both identifying actual heart disease cases (high recall) and making accurate positive predictions (high precision). This balance is often desirable in medical applications where both false positives and false negatives have significant costs.

3. Data Acquisition

The data used in this project is publicly sourced from the Kaggle dataset - Heart Failure Prediction Dataset.

The dataset was chosen as it combines data from 5 heart datasets over 11 common features, which makes it the largest heart disease dataset available so far for research purposes. The five datasets used for its curation are:

- Cleveland: 303 observations
- Hungarian: 294 observations
- Switzerland: 123 observations
- Long Beach VA: 200 observations
- Stalog (Heart) Data Set: 270 observations

4. Data Preprocessing/Preparation

The data processing and preparation was done through analysing the dataset, preparing the dataset for predictive modelling and defining the evaluation criteria for the prediction.

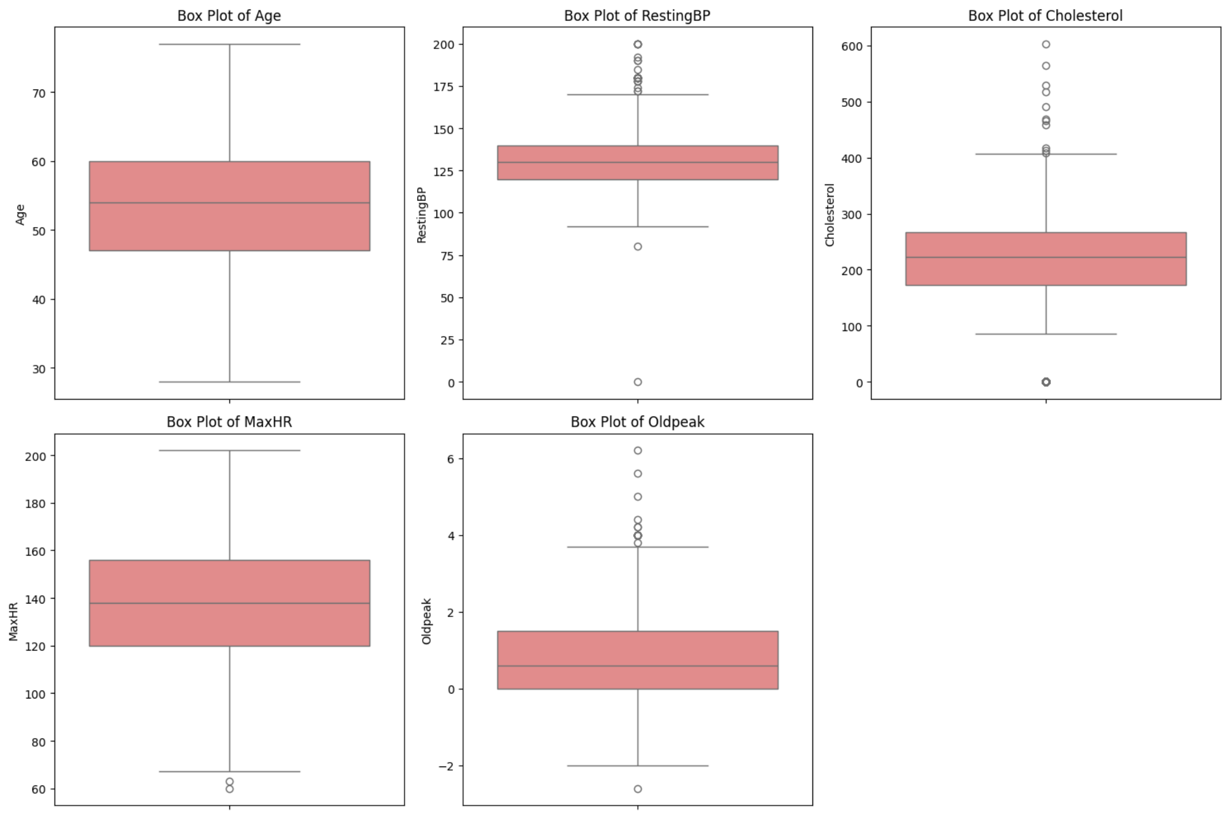
4.1 Data analysis

Below is a general analysis of the dataset

- The Dataset comprises 918 entries and 12 columns.
- The Data types are distributed among:
    - **7 numerical columns:**
        - Age
        - RestingBP
        - Cholesterol
        - FastingBS
        - MaxHR
        - HeartDisease
        - Oldpeak
    - **5 categorical columns:**
        - Sex
        - ChestPainType
        - RestingECG

- ExerciseAngina
- ST_Slope

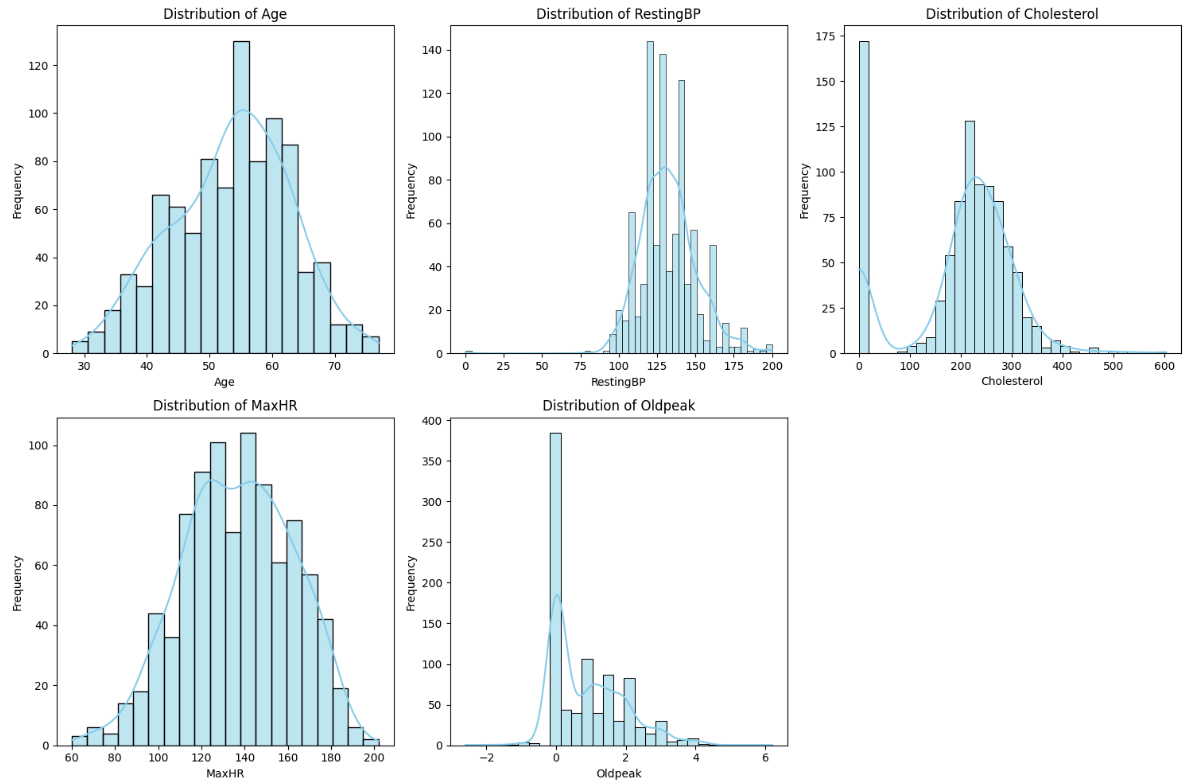
- There are no missing entries.
- There are no duplicates.
- **Outliers:**



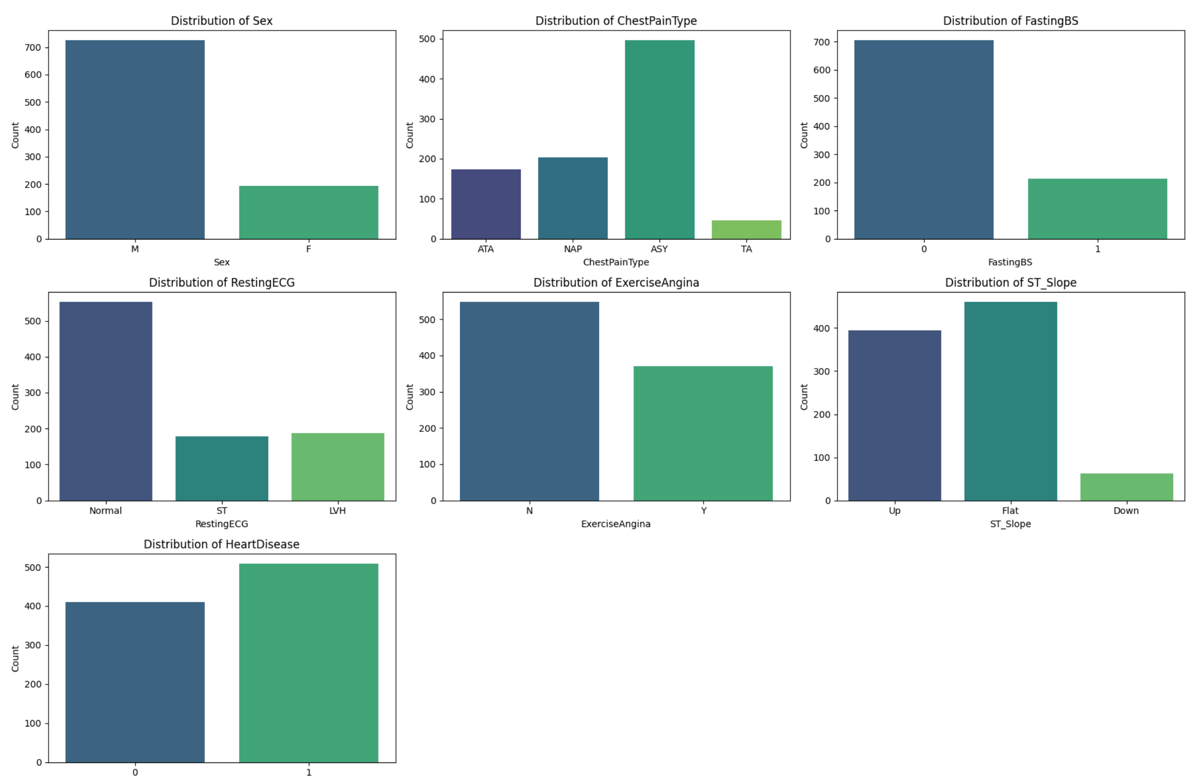
a.

- Based on the analysis of the features, the following observations were made regarding outliers:
  - Age: No significant outliers were detected.
  - RestingBP: 28 outliers were identified, with values falling below 90 or exceeding 170.
  - Cholesterol: 183 outliers were found, corresponding to values below 32 or above 407.
  - MaxHR: 2 outliers were noted, with values below 66 or above 210.
  - Oldpeak: 16 outliers were identified, with values below -2.25 or above 3.75.
  - Numerical features identified for outlier checking: ['Age', 'RestingBP', 'Cholesterol', 'MaxHR', 'Oldpeak']
  - Next Steps: Given that the objective is to determine the probability of heart diseases, it is expected that some patients in advanced stages of the condition will exhibit elevated values for Resting Blood Pressure (RestingBP), Cholesterol, and Oldpeak. **While these values may appear to be statistical outliers, they constitute valid recordings within the context of the problem statement and, consequently, should not be removed from the dataset.**

- Numerical feature distributions (visualized using histograms with KDEs), show varying patterns:
  - 'Age' and 'RestingBP' appear somewhat normally distributed.
  - 'Cholesterol' shows a bimodal-like distribution with a significant number of zero values.
  - 'Oldpeak' has a notable peak at zero.
  - 'MaxHR' shows a bell curve.
  - The plots are below:

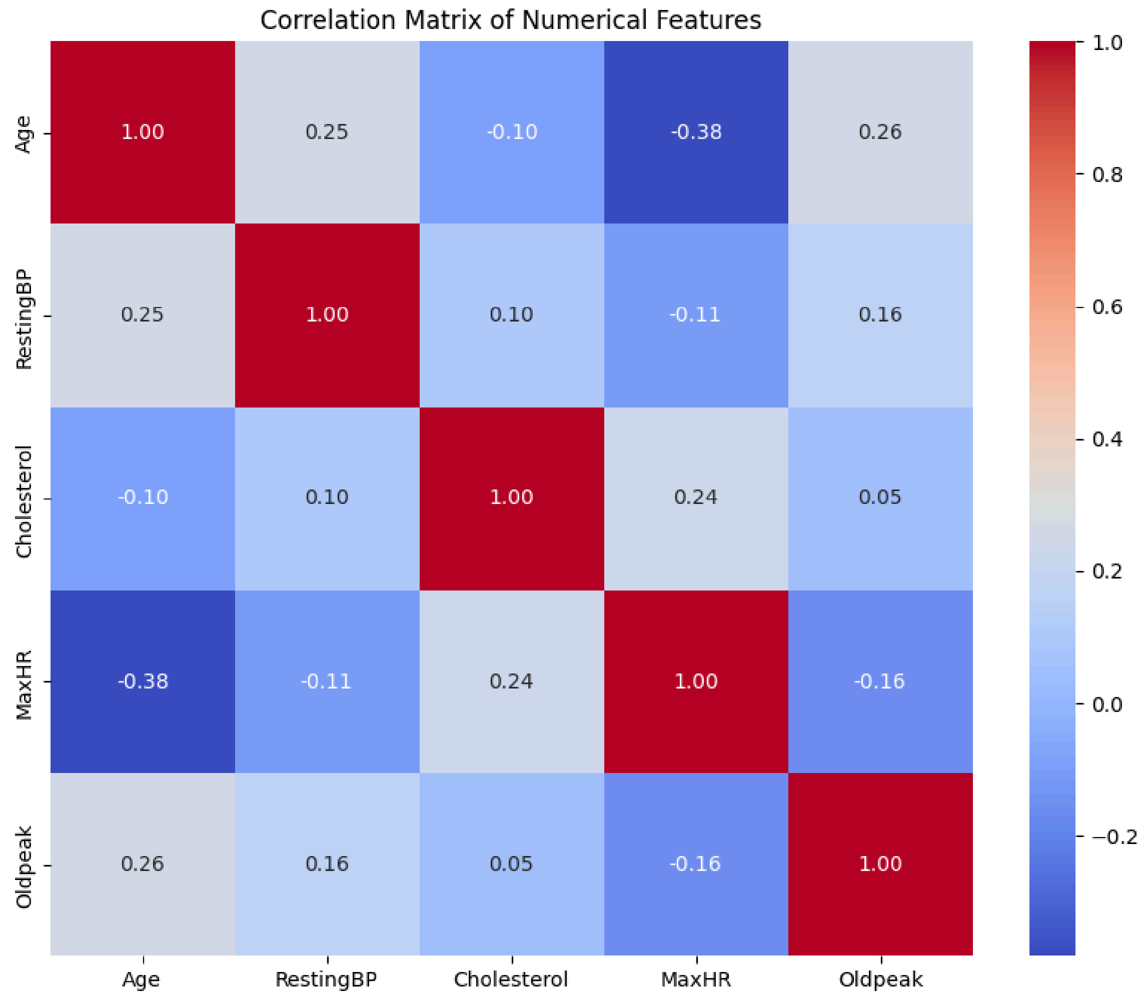


- Categorical feature distributions (visualized using bar plots), show varying patterns:
    a. Data set has more information about Male than Female that could lead to some bias in the prediction.
    b. 'FastingBS' shows that the majority of patients have a fasting blood sugar less than 120 mg/dl.
    c. 'HeartDisease' target variable appears relatively balanced, suggesting an even distribution between positive and negative cases, which is good for prediction.
    d. The plots are below:



- The correlation matrix of numerical features (visualized using a heatmap), indicates:
    a. 'Age' and 'MaxHR' have a moderate negative correlation (Logical as age increases, the person's MaxHR will decrease).

b. 'RestingBP' and 'Oldpeak' have weak positive correlations.



4.2 Data Preparation

- Feature engineering via Encoding
  - The "Categorical Columns" ('Sex', 'ChestPainType', 'RestingECG', 'ExerciseAngina', 'ST_Slope') are encoded using "One-hot encoding".
- Feature scaling via Normalization
  - The "Numerical Columns" are normalized using the standard scaler function from sklearn.
- Prepare for Supervised learning via splitting the entire DataFrame into:
  - The "Target Variable (y)" and assign the "Heart Disease" column to it.
  - The remaining columns as "Features (X)".
- Prepare for predictions
  - Split the data into "Training set" and "Testing Set" for future predictions with 80% of the data in the training set and 20% of the data in the testing set.

4.3 Evaluation Metrics

For the given data set, below are the key evaluation metrics that should be considered:

1. **F1-Score** - A high F1-score indicates that the model has good performance in both identifying actual heart disease cases (high recall) and making accurate positive predictions (high precision). **This balance is often desirable in medical applications where both false positives and false negatives have significant costs.**

Other Metrics that can also be considered are:

- **Accuracy** - A high accuracy metric provides an overview of the model's performance in predicting whether a patient has heart disease or not.
- **Precision** - A high precision means that when the model predicts a patient has heart disease, it is very likely that they actually do. This is crucial for avoiding unnecessary further diagnostic tests, treatments, or anxiety for patients who are incorrectly labeled as having heart disease (false positives).
- **Recall** - A high recall means the model is good at identifying nearly all patients who actually have heart disease. This is extremely critical in a medical diagnosis scenario like heart disease, as failing to identify a patient with the condition (a false negative) can lead to delayed treatment and severe health consequences.
- **AUC-ROC** - An AUC-ROC close to 1 indicates that the model is excellent at distinguishing between positive and negative classes across different thresholds. It tells us the probability that the model ranks a randomly chosen positive instance higher than a randomly chosen negative instance. In heart disease prediction, a high AUC-ROC means the model can effectively discriminate between healthy and diseased patients regardless of the chosen decision threshold, making it a reliable indicator of overall diagnostic capability.

5. Modeling

To train the model, key classification-based supervised modeling algorithms were selected such as:

- Logistic Regression
- Random Forest Classifier
- Decision Tree Classifier
- Support Vector Machine (SVC)
- Gradient Boosting Classifier
- Stacking Classifier (an ensemble method that combines the predictions of the other models)

To improve the model F1-Score results, the hyperparameters of these algorithms were tuned using GridSearchCV.

6. Model Evaluation

Both "Random Forest" and "Stacking Classifier" have relatively equal scores with a "high f1-score" score of 0.8920.

But given Random Forest has lesser computational complexity compared to Stacking Classifier (which needs to compute 5 methods and then run Logistic Regression on the result, leading to a total of 6 methods), **Random Forest would be a better model for this data set.**

Below is an overview of the model results by algorithm:

	Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
0	Logistic Regression	0.8370	0.8969	0.8131	0.8529	0.9142
1	Random Forest	0.8750	0.8962	0.8879	0.8920	0.9295
2	Decision Tree	0.8207	0.8700	0.8131	0.8406	0.9055
3	Gradient Boosting	0.8804	0.8972	0.8972	0.8972	0.9339
4	Support Vector Machine	0.8533	0.8704	0.8785	0.8744	0.9449
5	Stacking Classifier	0.8750	0.8962	0.8879	0.8920	0.9358

7. Front-end Interface

To make the application more useful for real-life applications, a front-end user interface was created using Streamlit and hosted on Streamlit community. The application can be accessed here.